

Project Design Phase

Solution Architecture

Date	26 June 2026
Team ID	Solo
Project Name	ShopEZ - E-Commerce Application
Maximum Marks	4 Marks

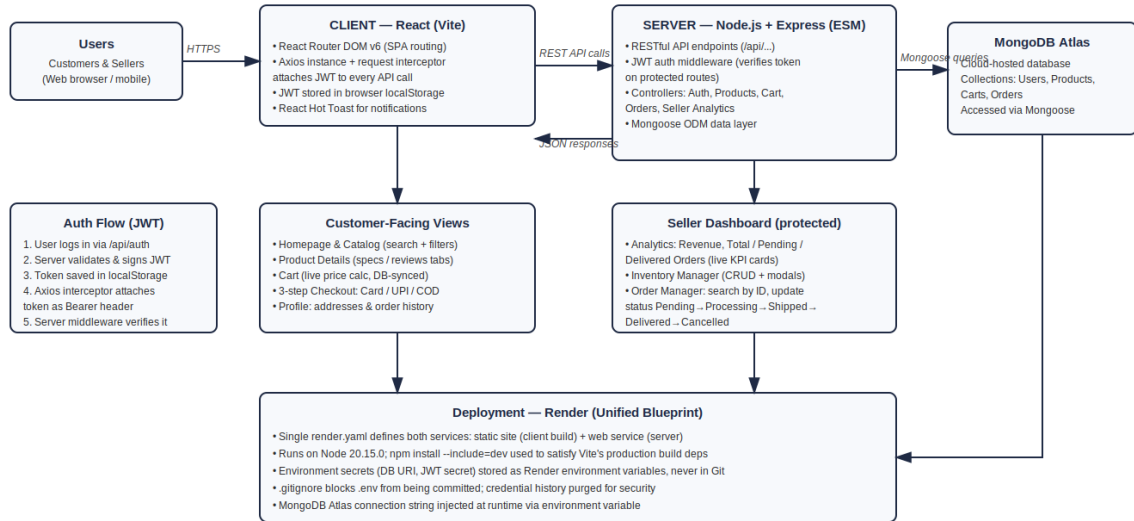
Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- ShopEZ follows a standard 3-tier MERN architecture. Users (customers and sellers) interact through a React (Vite) client, which uses React Router DOM v6 for navigation and an Axios interceptor to attach the JWT stored in localStorage to every request. This client talks to a Node.js/Express server (ES Modules) exposing RESTful API endpoints for authentication, products, cart, orders, and seller analytics, with middleware verifying JWTs on protected routes. The server uses Mongoose to read/write data in MongoDB Atlas, which stores Users, Products, Carts, and Orders collections.
- On the client side, customers get catalog browsing, product details, a live-synced cart, and a 3-step checkout (Card/UPI/COD), while sellers get a protected dashboard with live KPI analytics, inventory CRUD, and order-status management (Pending → Processing → Shipped → Delivered → Cancelled).
- The entire application is deployed on Render using a single render.yaml blueprint that defines both the static client build and the Express web service, running on Node 20.15.0. Environment secrets (DB connection string, JWT secret) are stored as Render environment variables and excluded from Git via .gitignore, keeping credentials out of the repository.

Example - Solution Architecture Diagram:

ShopEZ – Solution Architecture & Data Flow



Repository Structure:

Documentation/ — planning docs, requirement sheets, templates, PDF deliverables

Project/client/ — React frontend (compiled to static assets for deployment)

Project/server/ — Express server exposing RESTful API endpoints

.gitignore — excludes environment secrets (.env) from version control